

“The Role of AMRs and AGVs in Transforming Production Logistics: Applications, Safety, and Sustainability ”

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Abstract

This paper explores the use of Autonomous Mobile Robots (AMRs) and Automated Guided Vehicles (AGVs) in production logistics, focusing on their applications, safety features, and role in promoting sustainability. Through a review of academic papers and insights from study visits, the paper examines how these technologies can improve operational efficiency and reduce non-value-adding activities. Our findings show that AMRs and AGVs have great potential to enhance logistics systems, although integrating them with existing systems and ensuring real-time data synchronization are still key challenges. Additionally, the paper highlights the environmental benefits of these technologies, such as reduced energy consumption and emissions. Finally, the paper suggests areas for future research, including improving synchronization between digital systems and robots and exploring how these technologies can be scaled for broader industrial use.

Keywords: Autonomous Mobile Robots (AMRs), Automated Guided Vehicles (AGVs), Production logistics, Industrial automation, Sustainability, Real-time synchronization, System integration, Logistics automation.

1. Introduction

Automation is becoming an essential part of logistics in many industries, as it significantly impacts performance indicators related to safety, efficiency, cost, accuracy, and reliability. Industrial robots were first introduced in 1961 to target repetitive tasks that require a high level of precision, such as welding, painting, moving, and cutting. Mobile robots were later developed to integrate with production lines, primarily for transporting and delivering products. Unlike fixed robots or robotic manipulators, which operate in areas inaccessible to humans, autonomous mobile robots (AMRs) can interact with humans in the human environment and function as collaborative robots (cobots).

This research paper aims to discuss how AMRs operate in industrial environments for logistical activities, as well as to identify and highlight the advantages and disadvantages of AMRs compared to fixed robots, also known as manipulators, as observed during lab and site visits. Additionally, we aim to explore how "stock-management" activities can be optimized through the use of AMRs to improve both internal and external logistical processes within a production company.

To achieve these objectives, the paper is structured around the following research questions:

RQ1: *How do AMRs and Automated Guided Vehicles (AGVs) contribute to improving operational efficiency and reducing non-value-adding activities in production logistics?*

RQ2: *What are the primary challenges in integrating AMRs and AGVs with existing systems and ensuring real-time data synchronization?*

RQ3: *How can these technologies promote sustainability, including reduced energy consumption and emissions?*

RQ4: *What are the potential directions for future research and industrial scaling of AMRs and AGVs?*

The SCANIA Control Tower visit on November 7th serves as a great example of a "case study" for optimizing internal logistics

(stock management) using AMRs to minimize repetitive activities, which are later defined in this report as waste tasks that do not add value to the product.

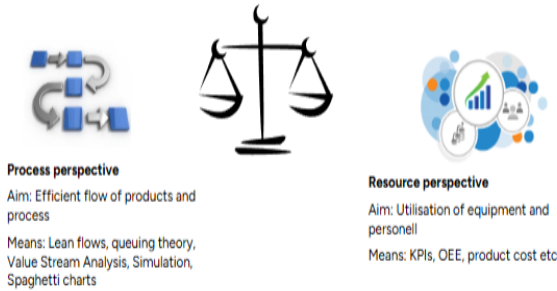
In terms of the paper's contributions, **Danny** will focus on comparing the capabilities and features of AMR and Automated Guided Vehicles (AGVs) in production logistics, specifically concerning safety, mission execution, and accuracy. **Nadir** will analyze the integration of AMR into production logistics systems, addressing issues such as layout, safety, and compatibility with legacy systems. Additionally, Nadir will further discuss the differences between AMRs and fixed robots based on insights from the Human-Robot Collaborative site visit. **Azar** will examine the reliability, flexibility, and cooperation of AMRs in material handling, as well as their integration with other systems, while also addressing ethical aspects and sustainability.

2. Methodology and Approach

2.1 Literature Review

For the literature review related to eliminating time wasted in non-value-adding tasks, the main method used was to follow the sequence of topics discussed during the lectures of ML2305 Production Logistics Supply Chain as well as ML2301 Production Management. Aligning topics discussed in both lectures and exploring scientific academic papers through Google Scholar that analyze titles related to performance measurement. Therefore, we undertook a literature review with keywords such as performance measurement, operations management, supply chain management and measurements, material flow measurement, and other similar terms related to material flow efficiency. The majority of results we got eventually directed us towards lean manufacturing subjects. Therefore, we decided to start with the early concepts that defined the need for automation through the elimination of waste, pioneered by the Toyota Production System (TPS) through

the Lean concept. Then we continued to explore more papers related to lowering non-value-adding activities. We focused on papers that showed results related to continuous improvement through lowering time and manpower wasted in non-value-adding activities through the use of automation in internal logistical activities. We connected the concepts in the papers to that of lecture 4 discussing the efficient flow of products and processes through Lean flows, queuing theory, and Value Stream Analysis. With an ultimate goal of using technology to reduce time, cost, and waste. We envisioned how fixed robots and AMRs can add flexibility and cooperation in lean flows and material handling tasks as shown below.



Expanding on this, the literature review of AMRs (Autonomous Mobile Robots) and AGVs (Automated Guided Vehicles) was conducted using well-established academic databases like Google Scholar, ResearchGate, IEEE Xplore, and ScienceDirect. These platforms are widely used in academic research, providing access to a variety of peer-reviewed papers, articles, and conference proceedings. Each source was carefully chosen based on how well it addressed our research questions. For example, we used Zhang et al. (2022) to understand how AGVs and AMRs are integrated into existing systems. This was important for answering questions about their compatibility with older infrastructures and how they improve logistics processes. Bechtsis et al. (2017) helped us explore the sustainability aspects of AGVs in supply chains, especially when it comes to their environmental impact. To dive deeper into the environmental concerns, we looked at Popescu et al. (2024), which discusses energy use and the environmental effects of robotics in logistics.

2.2 Search Strategy and Keywords Use

To find the right sources, we focused our keyword searches around the key aspects of this study: the capabilities and features of AMRs and AGVs, their integration into production logistics systems, performance in terms of reliability, flexibility, and cooperation in material handling tasks, as well as their sustainability impacts. We started by brainstorming terms and phrases related to our focus areas and then used them to search in the selected databases. For example, when comparing AMRs and AGVs, we searched using terms like “AMR vs AGV safety,” “AGV mission definitions,” and “AMR accuracy in logistics.” These helped us focus on studies that compared the two technologies in terms of safety features, mission execution, and overall accuracy. When looking into integration, we used keywords like “AMR integration with legacy systems,” “AGV layout optimization,” and “safety in automated logistics systems.” These keywords helped us find studies focused on how AMRs and AGVs fit into existing logistics systems and what challenges come with integrating them into older infrastructures. For performance-related questions, we used phrases like “reliability of AMRs,” “flexibility in AGVs,” and “cooperation between AGVs and AMRs in material handling tasks.” These keywords guided us toward research that explored how well these robots perform in dynamic environments and how they cooperate with human operators. Finally, for broader themes like sustainability and ethical considerations, we used terms such as “sustainability of robotics in logistics,” “ethical implications of

AMRs,” and “energy efficiency in AGVs.” These keywords helped us gather research on the long-term impacts of automation, including environmental sustainability and the social effects of automation technologies.

2.3 Search Strategy and Keywords Used

In identifying relevant studies, we used targeted keywords based on the key aspects of our research. For comparing AMRs and AGVs, we focused on terms like “AMR vs AGV safety,” “AGV mission definitions,” and “AMR accuracy in logistics,” to explore their differences in safety and performance. When examining integration, we used keywords such as “AMR integration with legacy systems” and “AGV layout optimization” to understand how these technologies fit into existing logistics infrastructures. For performance, we searched with phrases like “reliability of AMRs,” “flexibility in AGVs,” and “cooperation in material handling” to assess their functionality in dynamic environments. Finally, to explore sustainability and ethics, we used terms like “sustainability of robotics in logistics” and “energy efficiency in AGVs” to gather insights on the long-term environmental and social impacts of automation. To fine-tune our search and ensure we found the most relevant studies, we used Boolean operators like “AND” and “OR” to combine keywords and narrow down our results. For example, “AMR AND safety” helped us find studies specifically focused on the safety features of AMRs, while “AGV OR AMR flexibility” expanded the search to include studies that looked at both technologies' flexibility. We also applied filters such as “peer-reviewed” and “last 10 years” to ensure the relevance and credibility of the selected sources. This helped us stay focused on the most current research in the field.

3. State of the art

3.1 AMR's integration into production logistics systems

The earliest pioneer of waste elimination was the Toyota Production System. Through applying the JIT foundation they mastered the art of producing consistently without any waste. They build the theoretical foundation of continuous improvement philosophy that nowadays is called Lean. Lean is the westernization of a Japanese concept that has carried several names. It has variously been known as the Toyota Production System, JIT(Just in Time), Pull Manufacturing, TQM(Total Quality Management), and by various other names. Each of these names incorporates some aspects of lean failures. Continuous improvement, lean, and agile systems are becoming an essential part of every industrial logistics set-up. The end goal and main purpose are to reduce waste, as well as maximize customer value, and most importantly minimize ALL Non-value adding activities. The most common Non-value adding activity is “Internal-Transportation of material”. Transporting material internally is considered a repetitive task that has no direct or indirect added value towards the customer. This is precisely where AGVs or AMRs could play a key role through the reduction of such repetitive tasks which are considered non-value-adding tasks. AGVs or AMRs could enhance the logistical flow efficiency internally within the plant.

To enhance logistical flow efficiency it is important for AGV or AMR to fully recognize the design lay-out for logistics support. One of the methods that can be used to create a visualization of the plant layout is to create flow charts to understand the material flow between departments. Flow charts can present the layout and show the material flow directions and the fixed stations such as machining stations. Performance KPIs such as safety can be later measured and analyzed based on a well-designed flow path and layout of the manufacturing plant. There are various ways the theAGV/AMR flow path can be designed and implemented concerning the plant layout.

Other than safety, further performance checks such as “Transportation-time” can be further improved through investigations of the total number of AGVs or AMRs required to move certain amounts of material between stations or nodes. Therefore pathing or plant layout is of high importance for safety in production logistics systems when using AAGVs or AMRs.

AGV or A.M.R Autonomous Mobile Robots are developed and favored due to their autonomy and ability to position themselves and navigate without human intervention. AGVs or AMRs are unique and play an important role in the tasks of storage, transport, and production because they are independent and able to orient themselves in any manufacturing or production space without external guidelines. They can orient themselves using advanced level sensors and a virtual space map which makes them very flexible and independent.

Autonomous Guided Vehicles (AGVs) or Autonomous Mobile Robots (AMRs) are used within many different industries for automating internal transport of material. An AMR usually has built-in GPS and laser navigation systems that allow it to maneuver independently upon receiving data from synchronized cameras and sensors.

On the other hand, an AGV is currently considered a legacy system because AGV requires guidance in the form of sensors, wires, or magnetic strips that are built into the facility. The different vehicles receive an order, which is then processed and the material is delivered to the chosen destination. The process of picking up and delivering material varies and is often specifically designed for the process.



Figure 1: AGV and AMR

3.2 Comparing AMRs and AGVs: Mission, Accuracy, and Safety

Due to their different navigation systems, AGVs and AMRs have different capabilities and features regarding production logistics. AGVs are enhanced for high performance and predictability, while AMRs are intended for flexibility, agility, and simplicity. AGV features are based on predefined (virtual) paths and space discretization, so in many cases, AGVs support infrastructure. AMRs, on the other hand, exploit their smaller mobile base and onboard intelligence to aim for safe and flexible navigation in an infrastructure-free environment (ILIAD, 2024).

Layered architectures on AMR are a general design approach for navigation systems. The initial layered architecture was The Three-Layer Architecture introduced by David Kortenkamp in 1998, a hybrid reactive architecture that consists of three main blocks: (1) a reactive feedback control layer, (2) a reactive plan layer, and (3) a deliberative layer. This structure still exists and is commonly used in AMRs such as the Robotic Operating System (ROS) and ROS2 navigation, where a local planner runs at higher rates in parallel with a slower global planner. Every planner uses a private cost map to define the local environment, managing different semantics and thus using different rates and information resources (from sensors to objects or human trackers). Both ROS and ROS 2 navigation stacks are designed for single robots moving in dynamic unstructured environments (ILIAD, 2024).

3.2.1 Mission

A study conducted by Ziqiang Yu et al. (2013) designed a laser-guiding AGV control system consisting of hardware structure and software design. The hardware structure comprises a principal control system and an onboard control system. The onboard control system contains several modules such as control, communication, laser guiding & positioning, anti-collision, and driving & steering modules. AGV software design is responsible for controlling and driving the hardware. By its responsibilities, software quality, and intelligence system level depend on not only the algorithm itself but also the meaning of the system, which are software & hardware environment and program structure.

According to Ziqiang's design, the principal control system initiates to search for the idle AGVs in the system, picks the most effective and efficient one, and assigns the task information to it. Based on the AGV's current position and its mission. An optimal path is designed by the principal control system and the optimal path will be informed to the onboard control system by LAN wireless communication. After that, the onboard control obtains the position information by laser guiding & positioning module and a computer control unit calculates the corresponding control input based on some special control strategies. At the end, the driving & steering module receives the control input and performs it, ending with one control cycle.

The onboard system in Ziqiang's design is similar to the reactive feedback control layer, the principal control system adopts the reactive plan layer in three layers architecture, and the simulation system is a simplified version of a deliberative layer. As a complex computer control system, it is not sufficient to combine the needed hardware and software simply. The realization of autonomy depends on the software design to a great extent. Besides reforming and upgrading the intelligent hardware of the AGV, the software platform and program design have become one of the key factors for the AGV in realizing its scheduled aim. For instance, in the lab session, the AMR has modern hardware and sophisticated sensors but simple software operation. Furthermore, the reactive plan layer, which is the traffic management system, could not be demonstrated in the session because only one AMR was available in the lab.

3.2.2 Accuracy

A study conducted by Shi et al. in 2020 utilized Ultra Wide Band (UWB) technology to lower the cost of an AGV navigation system. The study used a gradient descent method and a least square method to achieve high-accuracy positioning. The result is very cost-efficient compared to the Real-Time Kinematic (RTK) satellite system. However, the study found that the layout of the UWB sensor affects the localization in a way that the robot gets better signal coverage in the interaction areas of the center regions of all sensors. For the application, a UWB receiver is installed to an AMR, and three other transmitters are stationed tens of meters apart and 1.5 meters high with a triangle position (see Figure 2). AMR receives range signals from three transmitters which share distance signals. Then, the system can cover an area of a couple of hundred square meters.

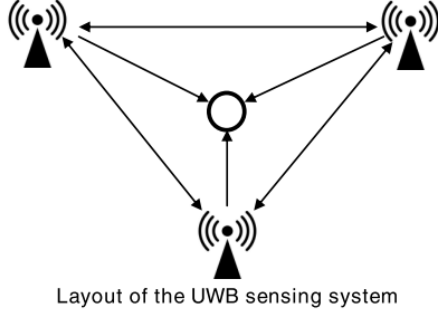


Figure 2 The middle circle represents a mobile robot equipped with a UWB receiver. Three others nearby are UWB transmitters.

In the KTH production logistic laboratory, the AMRs use the UWB navigation system for robot location, which is the same localization approach as Shi et al. (2020) study. Although in the logistic laboratory session, the AMRs had battery issues, which were not able to demonstrate the AMR's operation, the UWB location system was well demonstrated using the digital shadow. In the demonstration, the UWB system showed bad real-time tracking quality in some areas due to the transmitter deployment. It is aligned with the findings that the UWB layout is one of the factors influencing the performance of the UWB positioning system. However, having a low-cost and high-accuracy localization approach for AMRs is not enough without any purpose. For instance, Scania has shut down a research project that uses the UWB positioning system in the Södertälje Plant

3.2.3 Safety

Both AMR and AGV follow health and safety regulations (e.g., ISO 3691-4:2020) to account for general safety (e.g., collision avoidance) and perceived safety due to their operations in which humans and robots share the workspace, a process known as human-robot interaction (HRI). ILIAD (Intra-Logistics with Integrated Automatic Development, 2024) has developed safety layers in HRI (see in table X), which are: (1) safety stop – the minimum level of industry-standard safety certified laser scanner hardware, which avoids AGV/AMR from get near to any obstacle within a preconfigured safety zone, (2) vehicle-safe motion unit – assuring biomechanical safety of nearby people by velocity restriction using a mobile vehicle version of the generalized safe motion unit (vSMU), (3) situation awareness planning – classifying human-robot spatial interaction (HRSI) by encoding these in the form of cost maps, (4) navigation intent communication – comprising the two-way communication of navigation between robots and humans to improve perceived safety in HRI, (5) flow-aware motion planning – the top level safety layer using maps of dynamics (MoDs) that define human motion patterns.

No	Layer	Role
5	Flow-aware global motion planning	Create paths consistent with human patterns
4	Navigation intent communication	Improve the legibility of robot motions
3	Situation-aware constraints	Include local interactions in planning
2	Vehicle-safe motion unit	Limit robot speed to account for pain/injury biomechanics
1	Safety stops/ slowdowns	Avoid collisions

Table 1-HRI Safety Layers

In common industries, AGVs usually only fulfill the first safety layer and rarely meet the second layer requirements. For instance, Ziqiang Yu, et al, 2013 design demonstrates how the AGV systems

work only covered the second safety layer based on its onboard system, which has anti-collision, fault detection, laser positioning, and trajectory tracking.

Another example, the AMR that we have on the KTH Lab only qualifies the second layer with its capability to recognize biomechanically safe velocity limits obtained from live data about possibly colliding human body parts, inertial properties, and surface curvature for the mobile robot. In the third layer, KTH's AMR can generate a costmap around the human so that it can be used by other modules of the navigation architecture but, according to our Lab session, the AMR cannot fulfill the six HRSI classes which are: passing by on the left (PBL), passing by on the right (PBR), robot overtakes left (ROL), robot overtakes right (ROR), path crossing from the left (PCL), and path crossing from the right (PCR) (see Figure 3).

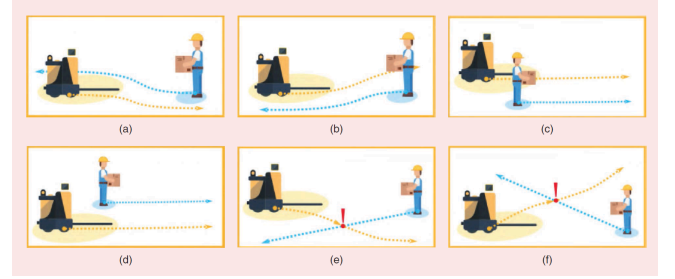


Figure 3 The six HRSI classes.

4. Observations

We visited the KTH Human-Robot Collab lab on 12th November 2024 at 9 am located at P202 Lab KTH Brinellv 68. We are grateful and thankful for a great informative session presented by "SICHAO". A detailed explanation of how ABB robotic arms complimented with 3 themes/demos of digital twins was presented. Also, a fully fitted with synchronized cameras and a well-functioning Automatic Mobile Robot was presented and described. Following the presentation, the use of digital twins in conjunction with the ABB robotic arms and the Automatic Mobile Robot particularly stood out. These digital twins accurately reflected the robots' activities, showcasing their potential to significantly enhance operational troubleshooting and workflow optimization. However, we noticed some challenges related to the real-time synchronization between the digital twins and the robots' actions. These delays highlight critical areas for future research, such as improving the response times and data processing capabilities of digital twin systems. Solving these issues could make these systems more dependable and effective in real-world industrial applications, improving both their predictiveness and efficiency.

5. Discussion

5.1 Evaluating Reliability, Flexibility, and Cooperation of AMRs and AGVs in Material Handling

As we have previously mentioned in the literature review, our goal is to eliminate time spent on non-value-adding tasks. Performance measurement is central to achieving this goal. It serves as a vital tool in optimizing the use of time for value-adding processes and ensuring the efficient flow of materials in and out of the conversion process. Timely flow, as defined, meets the needs of various organizational stakeholders, such as accounting (cash flow), marketing (consumer reaction times), and the overall steady-state flow of materials (sales and supply chain) (Afy-Shararah & Rich, 2018).

In line with these performance goals, the integration of automation technologies, such as Autonomous Mobile Robots (AMRs) and Automated Guided Vehicles (AGVs), can further enhance material flow and process efficiency. These robots are transforming material handling by automating tasks like transporting raw materials, managing inventory, and distributing finished goods. Their ability to integrate seamlessly with existing systems, including legacy ones, supports Lean principles by reducing manual handling and optimizing goods flow. This was demonstrated during our study visit to Scania, where AMRs and AGVs were integrated to improve internal logistics, driving both operational efficiency and sustainability (Larsson & Wernstrand, 2023).

Advanced communication technologies, particularly 5G, play a vital role in facilitating this integration. Research highlights that 5G networks provide the low-latency, high-speed data transfer required for real-time monitoring and control, even in environments with complex layouts and high material flow (Oyekanlu et al., 2020).

Building on this, 5G enhances fleet management by enabling real-time task allocation and the coordination of multiple robots. Centralized control systems leverage these capabilities to dynamically optimize workflows, reduce congestion, and minimize collision risks in high-traffic areas (Oyekanlu et al., 2020). These capabilities not only improve operational reliability but also reduce inefficiencies, reinforcing Lean's focus on continuous improvement and resource optimization.

Despite these advancements, achieving seamless integration poses challenges, particularly in environments relying on legacy systems. Bridging these gaps requires standardized protocols and interfaces to ensure compatibility between modern robotics and older infrastructure, minimizing disruptions during the transition (Oyekanlu et al., 2020).

Once these integration hurdles are addressed, the performance of AMRs and AGVs in material handling depends on three essential factors: reliability, flexibility, and cooperation. These attributes are critical for enabling the robots to handle the dynamic and often unpredictable demands of modern logistics environments, ensuring efficiency and adaptability.

5.1.2 Ensuring System Reliability in Material Handling

Reliability ensures that AMRs and AGVs can consistently perform their tasks without failure. This is especially crucial in material handling, where system interruptions can disrupt the entire logistics chain. By integrating 5G networks, organizations can significantly enhance system reliability through uninterrupted, low-latency communication. This technology enables real-time monitoring and control, even in high-pressure environments where timely data transmission is critical (Oyekanlu et al., 2020).

In addition to 5G, tools like Failure Modes and Effects Analysis (FMEA) play a proactive role in identifying and mitigating potential failure points. By analyzing possible failure modes, FMEA helps prevent system breakdowns, leading to improved overall resilience and minimizing downtime (Bekishev et al., 2023).

Together, these technologies and methodologies strengthen the reliability of material handling operations.

5.1.3 Adapting to Dynamic Environments: Testing Flexibility

While reliability ensures stability, flexibility allows AMRs and AGVs to adapt to the ever-changing demands of modern logistics. In modern material handling, where inventory levels and warehouse layouts frequently change, flexibility enables continuous operations. AMRs equipped with AI-driven path-planning algorithms can autonomously adjust their routes in response to real-time obstacles, ensuring uninterrupted task

execution (Hercik et al., 2022). To illustrate this, *Figure 2* shows how mobile robots navigate by using real-time data to adjust their paths. This helps them optimize routes and adapt better to complex environments (Hercik et al., 2022).

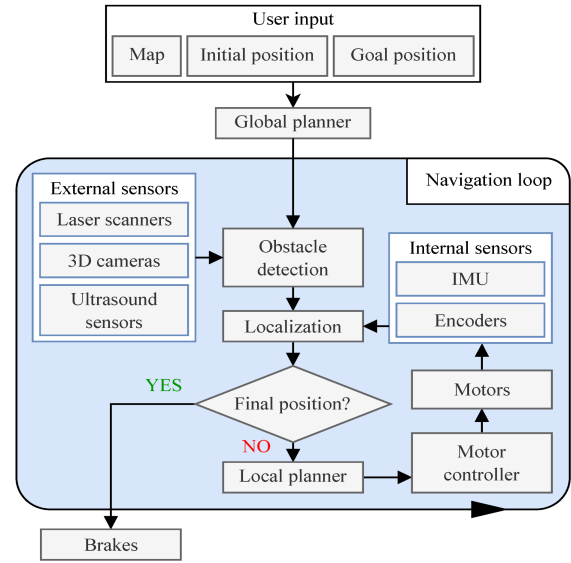


Figure 2: Navigation principle of mobile robots

Task versatility, a key aspect of flexibility, further enhances the ability of AMRs to adapt to varied tasks. By seamlessly transitioning between different operational roles, AMRs can optimize workflows and improve overall efficiency in dynamic environments, such as smart factories (Larsson & Wernstrand, 2023). This adaptability ensures that AMRs remain effective even as operational demands shift.

5.1.4 Enhancing Cooperation in Shared Workspaces

In addition to reliability and flexibility, cooperation is crucial for maximizing the potential of AMRs and AGVs in shared workspaces. Effective cooperation allows these systems to work harmoniously with each other and with human operators, optimizing workflows and preventing bottlenecks. Cooperation relies on robust communication frameworks, such as sensor fusion, which enable synchronized task execution and real-time data sharing. This coordination minimizes bottlenecks and enhances throughput, even in high-traffic areas (Larsson & Wernstrand, 2023). Centralized traffic management systems further support cooperation by dynamically assigning tasks and rerouting robots to avoid congestion (Oyekanlu et al., 2020).

By focusing on reliability, flexibility, and cooperation, organizations can unlock the full potential of AMRs and AGVs in material handling. These attributes enable the robots to meet the complex and dynamic demands of modern logistics environments, driving efficiency and productivity to new heights.

5.2 Applications, Integration, and Sustainability of AMRs and AGVs

Following the discussion on operational performance, it's essential to examine how AMRs and AGVs extend their impact across various domains.

Beyond enhancing reliability, flexibility, and cooperation, these technologies demonstrate their value through diverse applications, seamless integration with existing systems, and their role in addressing ethical and sustainability challenges.

5.2.1 Use Cases of AMRs and AGVs

AMRs and AGVs have proven their versatility in numerous industries, each application showcasing their adaptability and impact. In logistics and warehousing, they automate labor-intensive tasks such as order picking and inventory management, significantly improving efficiency and reducing manual labor (Popescu et al., 2024).

Similarly, in manufacturing, they streamline workflows by transporting materials between workstations, minimizing delays, and optimizing production schedules (Costa et al., 2022).

In addition to these traditional sectors, AMRs are increasingly deployed in healthcare, where they assist with drug delivery and patient mobility, enhancing both operational efficiency and patient outcomes (Zhang et al., 2023). The construction industry also benefits from their use in site inspections and heavy material transportation, improving safety and reducing physical strain on workers (Zhang et al., 2023). These varied use cases highlight their ability to adapt to the unique demands of different environments.

5.2.2 Integration with Other Systems

The success of AMRs and AGVs in these diverse applications relies heavily on their ability to integrate seamlessly into existing system architectures. By connecting with Enterprise Resource Planning (ERP) and Warehouse Management Systems (WMS), these robots facilitate real-time data exchange and dynamic task allocation, ensuring coordinated and efficient operations (Costa et al., 2022).

The Internet of Things (IoT) further enhances this integration by enabling continuous communication between robots, sensors, and control systems.

This interconnectedness supports advanced functionalities such as optimized route planning and collision prevention, improving overall system efficiency (Bechtsis et al., 2017). Additionally, integration with energy management systems allows for smarter battery usage, reducing both energy consumption and operational costs (Popescu et al., 2024).

5.2.3 Ethical Aspects

In the broader context of industrial transformation, the integration of AMRs and AGVs introduces several ethical considerations, particularly regarding their impact on employment. While these technologies automate repetitive and labor-intensive tasks, they also create opportunities for workers to transition into higher-skilled roles, such as system monitoring and robot maintenance.

This shift emphasizes technical skills and supports workforce upskilling, enabling organizations to adapt to evolving technological landscapes (Bechtsis et al., 2017).

These ethical considerations highlight the importance of fostering a balanced approach, where technological innovation aligns with human and organizational well-being. By proactively addressing these challenges, companies can establish an ethical framework that supports both technological progress and workforce development.

5.2.4 Sustainability

In addition to addressing ethical considerations, AMRs and AGVs play a pivotal role in advancing sustainability goals, emphasizing the alignment of technological progress with environmental responsibility. Their energy-efficient designs, often powered by renewable energy sources, help reduce greenhouse gas emissions compared to traditional fuel-based systems (Bechtsis et al., 2017). Advanced energy management systems further enhance efficiency by optimizing battery usage and incorporating energy recovery technologies, such as capturing energy during braking, to

minimize consumption (Popescu et al., 2024).

Sustainability extends to the lifecycle of these robots as well. Many organizations are adopting circular economy principles, focusing on recycling and reusing components to reduce electronic waste and lower production costs (Popescu et al., 2024). Furthermore, by optimizing logistics and minimizing unnecessary travel, AMRs and AGVs help create more sustainable supply chains, reducing emissions and conserving resources. These efforts not only support environmental goals but also reinforce the economic viability of adopting these technologies.

6. Conclusion

In this paper, we explored the integration of Autonomous Mobile Robots (AMRs) and Automated Guided Vehicles (AGVs) within production logistics, with a particular focus on their applications, safety features, and sustainability aspects. Our analysis reveals that both AMRs and AGVs have the potential to significantly enhance operational efficiency, flexibility, and reliability within logistics systems. Additionally, these technologies contribute to more sustainable practices by reducing non-value-adding activities.

However, while both technologies offer unique advantages, integrating them into existing systems presents some challenges. One of the biggest obstacles is ensuring smooth interoperability with legacy technologies, which is essential for realizing the full potential of these robots. The use of IoT and 5G technologies could help improve real-time data exchange and task allocation, but there are still issues with system compatibility that need to be worked through.

From a sustainability standpoint, both AMRs and AGVs offer clear benefits. They help reduce energy consumption, minimize emissions, and support circular economy principles, such as the recycling and reuse of components. As industries continue to adopt these technologies, they not only improve operational efficiency but also contribute to broader environmental goals, reinforcing the role of AMRs and AGVs in sustainable production and logistics.

While the results from our study are promising, there are several areas that still require further investigation. In particular, the real-time synchronization challenges between digital twins and robots, as we observed during the KTH Human-Robot Collaborative site visit, remain an important issue that needs to be addressed.

Additionally, more research is needed to understand how these systems can scale effectively and remain reliable in high-demand environments.

In conclusion, AMRs and AGVs are poised to transform production logistics by offering solutions that improve efficiency, flexibility, and sustainability. Although there are challenges to overcome, such as integration and synchronization, ongoing research and development will likely lead to more widespread adoption and improved performance in various industrial applications.

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